

Multimodal Context Mining for Semantic Labeling of Industrial Images

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Motivation

Limitations & Research Gap

- Image-based detection and coordinate-centric labeling identify object locations but **ignore object semantics**
- Visually similar objects may have different semantics, leading to **visual-semantic ambiguity**
- Underused context (text, lit state, surroundings) leaves a persistent **semantic judgment burden**

The Need for Context-Aware Semantic Labeling

- Reduce the semantic judgment burden by allowing annotators to **confirm semantic candidates** instead of interpreting every object individually.
- A single confirmed meaning can be **propagated to all objects** sharing the same semantics, enabling practical labeling from a cold-start

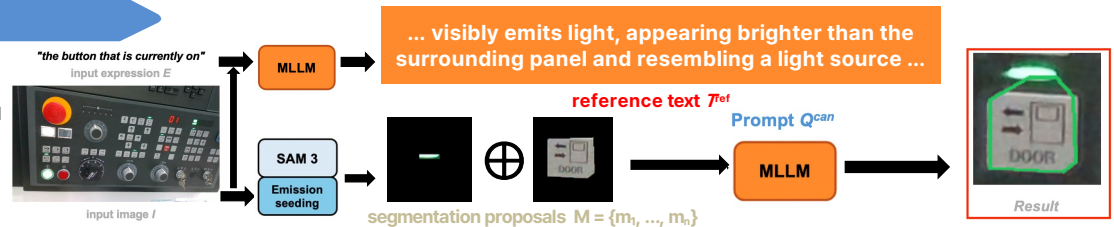
We propose Multimodal Context Mining for semantic candidate generation and label propagation

이 논문은 정부(교육부)의 재원으로 한국산업기술진흥원의 지원(P0030395, 첨단산업인재양성부트캠프)과 2026 정부(과학기술정보통신부)의 지원을 받아 수행된 연구임 (RS-2026-25490215)

Methodology

State Recognition via Attribute Prompting:

Button-lamp states are determined via RESAnything-based attribute prompting and pixel-level validation.



Concept Segmentation : SAM 3 PCS to segment all matching objects in a single pass.

"gray button"
Text prompt



Image exemplar

Exemplars - three clicks



All matching objects are segmented in a single pass



Output for image i



Propagation to all data

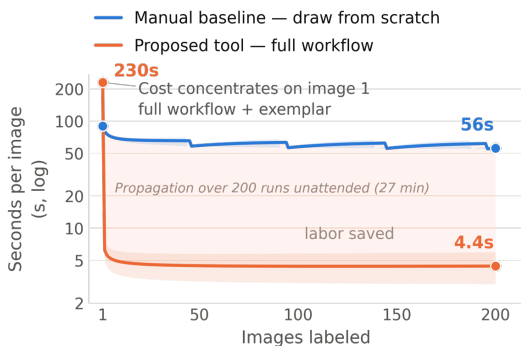
Semantic candidates Generation:

- OCR Extraction:** Extract text from industrial images
- Multimodal Prompting:** Combine OCR, image, and class context
- Candidate Generation:** Generate semantic candidates using MLLM
- Dictionary Matching:** Refine candidates via edit distance ($d \leq 2$)
- Concept Prompting:** Use refined candidates as SAM 3 prompts

Main Experiment

72 s/image → 230 s once + confirmation

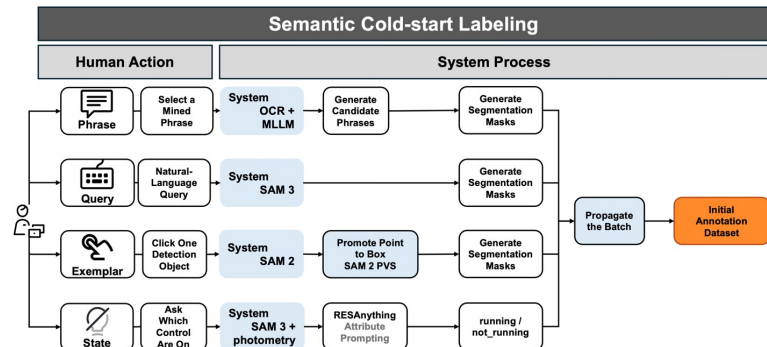
200 images: 4.3 h → 19 min



Human-in-the-Loop

Semantic Cold-Start Labeling:

We provide **four semantic labeling options**, enabling users to specify object meaning through different inputs and segment all matching objects in a single pass with SAM 3.



Conclusion & Future Work

- From coordinates to meaning:** Building on our multimodal context mining framework, we demonstrated the feasibility of efficient semantic labeling and aim to further advance meaning-centric annotation.
- UI/UX for non-experts:** Building on our semantic labeling framework, we aim to redesign the interface with self-orchestrating AI agents, enabling intuitive annotation for non-experts while improving efficiency and accuracy.